

CSS Print Media

Complete Guide: From Basics to Advanced

CSS Guide

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Chapter 1

Introduction

1.1 What is CSS Print Media?

CSS print media allows you to apply different styles when a webpage is printed or viewed in print preview mode. The `@media print` rule enables developers to create optimized, printer-friendly versions of web content without modifying the HTML or creating separate pages.

```
/* Basic syntax */
@media print {
    /* Styles applied only when printing */
}

/* Practical example */
@media print {
    body { font-size: 12pt; }
    .no-print { display: none; }
}
```

Print styling is crucial for creating professional documents, reports, invoices, and any content users might want to preserve on paper.

1.2 Why Print Styling Matters

1.2.1 Real-World Use Cases

- **Invoices & Receipts:** Professional formatted billing documents
- **Articles & News:** Clean, readable printed articles
- **Reports & Data:** Organized data displays for printing
- **Tickets & Reservations:** Printable confirmation documents
- **Certificates:** Professional-looking award documents
- **Legal Documents:** Properly formatted contracts and forms

1.2.2 Benefits of Print Styling

- Remove navigation, ads, and UI elements not needed on paper
- Adjust colors, fonts, and spacing for print readability
- Control page breaks and orphans/widows
- Reduce paper waste with optimized layouts
- Improve user experience for document printing
- Professional appearance of printed content

1.3 Browser Support

Print media styling is universally supported across all browsers and has been for decades:

Browser	Status
Chrome	Full Support
Firefox	Full Support
Safari	Full Support
Edge	Full Support
Opera	Full Support
Internet Explorer	Full Support
All Mobile Browsers	Full Support

Chapter 2

Core Concepts

2.1 Understanding Print Media Queries

2.1.1 Basic Syntax

```
/* Styles applied only when printing */
@media print {
    /* All CSS properties work normally */
    body {
        font-size: 12pt;
        color: black;
        background: white;
    }
}
```

2.1.2 Combining with Other Media Features

```
/* Print only on color printers */
@media print and (color) {
    .color-graphic { display: block; }
}

/* Print only on monochrome printers */
@media print and (monochrome) {
    .color-graphic { display: none; }
}

/* Print only on specific page orientation */
@media print and (orientation: landscape) {
    body { margin: 0.5in; }
}

/* Print on a minimum page width */
@media print and (min-width: 400mm) {
    .wide-layout { display: block; }
}
```

2.2 Print-Specific CSS Properties

2.2.1 Page Properties

The `@page` rule controls page margins, size, and orientation:

```
/* Set page size to A4 with 1-inch margins */
@page {
    size: A4;
    margin: 1in;
}

/* Set page to landscape */
@page {
    size: letter landscape;
    margin: 0.5in;
}

/* Different margins for first page */
@page :first {
    margin-top: 2in;
}

/* Named pages for different sections */
@page chapter {
    margin-bottom: 1.5in;
}

.chapter {
    page: chapter;
}
```

2.2.2 Page Margin Boxes

```
/* Add header and footer to every page */
@page {
    @top-center {
        content: "Document Title";
    }

    @bottom-center {
        content: "Page " counter(page) " of " counter(pages);
    }

    @top-left {
        content: string(title);
    }
}
```

2.3 Common Print Properties

2.3.1 Page Break Control

```
/* Force page break before element */
.new-section {
    page-break-before: always;
}

/* Force page break after element */
.section-end {
    page-break-after: always;
}

/* Avoid page break inside element */
.keep-together {
    page-break-inside: avoid;
}

/* Modern equivalent */
.new-section {
    break-before: page;
}

.section-end {
    break-after: page;
}

.keep-together {
    break-inside: avoid;
}
```

2.3.2 Widow and Orphan Control

```
/* Widow: Last line of paragraph on new page */
p {
    widow: 3; /* Keep at least 3 lines at top */
}

/* Orphan: First line of paragraph at bottom */
p {
    orphan: 3; /* Keep at least 3 lines at bottom */
}
```


Chapter 3

Practical Examples

3.1 Example 1: Article Printing

```
/* Default screen styles */
body {
    font-family: -apple-system, BlinkMacSystemFont,
        'Segoe UI', Roboto, sans-serif;
    font-size: 16px;
    line-height: 1.6;
}

header, footer, nav, .sidebar, .ads {
    display: block;
}

/* Print styles */
@media print {
    /* Remove non-essential elements */
    header, footer, nav, .sidebar, .ads,
    .comments, .share-buttons {
        display: none !important;
    }

    /* Optimize readability */
    body {
        font-size: 12pt;
        line-height: 1.5;
        color: black;
        background: white;
    }

    /* Article container */
    article {
        max-width: none;
        margin: 0;
    }

    /* Heading styles */
    h1 {
```

```
        font-size: 24pt;
        margin-top: 0;
        page-break-after: avoid;
    }

    h2 {
        font-size: 18pt;
        margin-top: 20pt;
        page-break-after: avoid;
    }

    h3 {
        font-size: 14pt;
        page-break-after: avoid;
    }

    /* Keep paragraphs with headings */
    h2 + p, h3 + p {
        page-break-before: avoid;
    }

    /* Image handling */
    img {
        max-width: 100%;
        height: auto;
        page-break-inside: avoid;
    }

    /* Links */
    a {
        color: black;
        text-decoration: none;
    }

    /* Show URL for external links */
    a[href]:after {
        content: " (" attr(href) ")";
        font-size: 0.8em;
        color: #666;
    }

    /* Avoid printing internal anchors */
    a[href^="#"]:after {
        content: "";
    }
}
```

3.2 Example 2: Invoice Printing

```
/* Screen styles */
.invoice {
    background: #f5f5f5;
}
```

```
border: 1px solid #ddd;
border-radius: 8px;
}

.action-buttons {
  background: #667eea;
  color: white;
  padding: 20px;
  text-align: center;
}

/* Print styles */
@media print {
  /* Page setup */
  @page {
    size: letter;
    margin: 0.5in;
  }

  /* Hide interactive elements */
  .action-buttons, .print-button,
  .edit-button, .delete-button {
    display: none !important;
  }

  /* Invoice container */
  .invoice {
    background: white;
    border: none;
    border-radius: 0;
    box-shadow: none;
    page-break-after: always;
  }

  /* Header */
  .invoice-header {
    border-bottom: 2px solid black;
    padding-bottom: 20pt;
    margin-bottom: 20pt;
  }

  .company-logo {
    max-width: 2in;
    height: auto;
  }

  .invoice-title {
    font-size: 18pt;
    font-weight: bold;
    margin: 10pt 0;
  }

  /* Invoice details */
  .invoice-details {
```

```
        display: grid;
        grid-template-columns: 1fr 1fr;
        margin-bottom: 20pt;
        font-size: 10pt;
    }

    /* Line items table */
    .invoice-items {
        width: 100%;
        border-collapse: collapse;
        margin: 20pt 0;
    }

    .invoice-items thead {
        border-bottom: 2pt solid black;
    }

    .invoice-items th {
        text-align: left;
        padding: 10pt;
        font-weight: bold;
    }

    .invoice-items td {
        padding: 8pt 10pt;
        border-bottom: 1pt solid #ddd;
    }

    .invoice-items tfoot {
        border-top: 2pt solid black;
        font-weight: bold;
    }

    /* Totals section */
    .invoice-totals {
        text-align: right;
        margin-top: 20pt;
        font-size: 11pt;
    }

    .total-amount {
        font-size: 14pt;
        font-weight: bold;
        margin-top: 10pt;
    }

    /* Footer */
    .invoice-footer {
        margin-top: 40pt;
        padding-top: 20pt;
        border-top: 1pt solid #ddd;
        font-size: 9pt;
        color: #666;
    }
```

```
/* Keep amounts together */
.invoice-items, .invoice-totals {
    page-break-inside: avoid;
}
}
```

3.3 Example 3: Report with Multiple Pages

```
@media print {
    /* Page setup */
    @page {
        size: letter portrait;
        margin: 1in;
    }

    @page :first {
        margin-top: 1.5in;
    }

    /* Remove navigation */
    nav, .sidebar, .breadcrumbs {
        display: none;
    }

    /* Title page */
    .report-cover {
        page-break-after: always;
        text-align: center;
        padding: 3in 0;
    }

    .report-title {
        font-size: 28pt;
        font-weight: bold;
        margin-bottom: 20pt;
    }

    .report-date {
        font-size: 12pt;
        color: #666;
    }

    /* Table of contents */
    .toc {
        page-break-after: always;
    }

    .toc h2 {
        font-size: 18pt;
        page-break-after: 10pt;
    }
}
```

```
.toc ul {
    list-style: none;
    padding: 0;
}

.toc li {
    margin-bottom: 8pt;
}

.toc a:after {
    content: leader('.') ' ' target-counter(attr(href), page);
}

/* Section breaks */
.section {
    page-break-before: always;
    padding-top: 0;
}

.section:first-of-type {
    page-break-before: avoid;
}

/* Chapter headings */
h1 {
    font-size: 24pt;
    margin: 30pt 0 15pt 0;
    page-break-after: avoid;
    page-break-before: always;
    border-bottom: 2pt solid black;
    padding-bottom: 10pt;
}

h2 {
    font-size: 16pt;
    margin: 20pt 0 10pt 0;
    page-break-after: avoid;
}

/* Figures and captions */
figure {
    page-break-inside: avoid;
    text-align: center;
    margin: 20pt 0;
}

figcaption {
    font-style: italic;
    font-size: 10pt;
    margin-top: 10pt;
}

/* Tables */
```

```
table {  
    border-collapse: collapse;  
    width: 100%;  
    margin: 15pt 0;  
}  
  
table, thead, tbody, tr {  
    page-break-inside: avoid;  
}  
}
```

3.4 Example 4: Receipt Printing

```
@media print {  
    /* Thermal printer width (80mm) */  
    @page {  
        size: 80mm auto;  
        margin: 0;  
    }  
  
    body {  
        width: 80mm;  
        margin: 0;  
        padding: 0;  
        font-family: monospace;  
        font-size: 10pt;  
    }  
  
    .receipt {  
        margin: 0;  
        padding: 0;  
        width: 100%;  
        text-align: center;  
    }  
  
    .receipt-header {  
        padding: 10mm 5mm;  
        border-bottom: 2pt solid black;  
        font-weight: bold;  
    }  
  
    .receipt-items {  
        padding: 10mm 5mm;  
        text-align: left;  
    }  
  
    .item-row {  
        display: flex;  
        justify-content: space-between;  
        margin: 5pt 0;  
        font-size: 9pt;  
    }  
}
```

```
.receipt-total {  
    padding: 10mm 5mm;  
    border-top: 2pt solid black;  
    border-bottom: 2pt solid black;  
    font-weight: bold;  
    font-size: 12pt;  
    text-align: right;  
}  
  
.receipt-footer {  
    padding: 10mm 5mm;  
    font-size: 8pt;  
    color: #666;  
    text-align: center;  
}  
}
```


Chapter 4

Advanced Techniques

4.1 Page Numbering

```
@media print {
  @page {
    @bottom-right {
      content: "Page " counter(page) " of " counter(pages);
    }
  }

  /* Reset counter for new section */
  .new-section {
    page-break-before: always;
    counter-reset: page 1;
  }
}
```

4.2 Color Handling

```
@media print {
  /* Force black text */
  body {
    color: black;
    background: white;
  }

  /* Convert colored text to black */
  a {
    color: black;
    text-decoration: underline;
  }

  /* Keep important colors for charts */
  .chart-legend span {
    color: inherit;
  }
}
```

```
/* For color printers */
@media (color) {
    .color-important {
        color: #667eea;
    }
}

/* For monochrome printers */
@media (monochrome) {
    .color-important {
        color: black;
        font-weight: bold;
    }
}
}
```

4.3 Print-Only Content

```
/* Hide from screen, show in print */
.print-only {
    display: none;
}

@media print {
    .print-only {
        display: block;
    }
}

/* Hide from print, show on screen */
.screen-only {
    display: block;
}

@media print {
    .screen-only {
        display: none !important;
    }
}

/* Example: Print QR code with instructions */
.print-only {
    text-align: center;
    margin-top: 30pt;
    padding: 20pt;
    border-top: 1pt solid #ddd;
}

.print-only-instruction {
    font-size: 10pt;
    color: #666;
}
```

Chapter 5

Best Practices

5.1 Key Recommendations

5.1.1 1. Always Test Print Preview

Before deploying, test in:

- Chrome Print Preview (Ctrl+P)
- Firefox Print Preview (Ctrl+P)
- Safari Print Preview (Cmd+P)
- Actual printer output
- Different page sizes (Letter, A4, Legal)

5.1.2 2. Use Absolute Sizes

```
/* Good: Absolute sizes for print */
@media print {
  body { font-size: 12pt; }
  h1 { font-size: 24pt; }
}

/* Avoid: Relative sizes can be unpredictable */
@media print {
  body { font-size: 1.2em; }
}
```

5.1.3 3. Hide Interactive Elements

```
@media print {
  /* Hide UI elements */
  nav, .sidebar, button, input,
  .advertisement, .print-button {
    display: none !important;
  }
}
```

```
/* Show essential information */
.print-only {
    display: block;
}
}
```

5.1.4 4. Control Page Breaks

```
@media print {
    /* Keep headings with content */
    h2, h3 {
        page-break-after: avoid;
    }

    /* Prevent orphans/widows */
    p {
        orphan: 3;
        widow: 3;
    }

    /* Keep tables together */
    table {
        page-break-inside: avoid;
    }
}
```

5.1.5 5. Optimize Colors

```
@media print {
    /* White background saves ink */
    body {
        background: white;
    }

    /* Dark text prints better */
    body {
        color: black;
    }

    /* Remove background images (use borders instead) */
    .box {
        border: 1pt solid black;
        background: white;
    }
}
```

5.2 Common Patterns

Pattern	Use Case
Hide navigation	Every print stylesheet
White background	Save ink and improve readability
Control page breaks	Multi-page documents
Show URLs for links	Articles and references
Page numbering	Long documents

Chapter 6

Common Issues & Solutions

6.1 Issue 1: Content Being Cut Off

Problem: Important content disappears in print.

Solution:

```
@media print {  
  /* Set appropriate margins */  
  body { margin: 0.5in; }  
  
  /* Ensure content fits */  
  .container { max-width: 100%; }  
  
  /* Use page-break-inside: avoid */  
  .important-section {  
    page-break-inside: avoid;  
  }  
}
```

6.2 Issue 2: Colors Not Printing

Problem: Colors appear different or don't print.

Causes:

- Printer in grayscale mode
- Background colors set but text not visible
- Color-only information without fallback

Solution:

```
@media print {  
  /* Use dark text for readability */  
  body { color: black; }  
  
  /* Provide fallback for monochrome */  
  @media (monochrome) {  
    .color-text {
```

```
        color: black;
        font-weight: bold;
    }
}
```

6.3 Issue 3: Excessive Page Breaks

Problem: Too many blank pages or unwanted breaks.

Solution:

```
@media print {
    /* Only break before major sections */
    .section { page-break-before: always; }

    .section:first-of-type {
        page-break-before: avoid;
    }

    /* Avoid breaking inside content */
    .keep-together {
        page-break-inside: avoid;
    }
}
```

6.4 Issue 4: Headers and Footers

Problem: Can't add headers/footers in CSS.

Note: The @page rule and margin boxes have limited support. Some browsers don't support them.

Workaround:

```
@media print {
    /* Fixed header (may not print on every page) */
    .header {
        position: fixed;
        top: 0;
        width: 100%;
        border-bottom: 2pt solid black;
    }

    /* Add space for header */
    body { margin-top: 60pt; }
}
```

Chapter 7

Summary & Quick Reference

7.1 When to Use Print Media

Use print media styling when:

- Creating printable documents
- Building invoice/receipt systems
- Providing article printing
- Generating reports
- Creating certificates or tickets

7.2 Quick Reference

7.2.1 Basic Print Stylesheet

```
@media print {  
  /* Remove interactive elements */  
  nav, .sidebar, .ads, button {  
    display: none !important;  
  }  
  
  /* Optimize typography */  
  body {  
    font-size: 12pt;  
    line-height: 1.5;  
    color: black;  
    background: white;  
  }  
  
  /* Control page breaks */  
  h2, h3 {  
    page-break-after: avoid;  
  }  
  
  table {
```



```
    page-break-inside: avoid;
  }

  /* Show URLs */
  a:after {
    content: " (" attr(href) ")";
  }
}
```

7.2.2 Common Print Properties

Property	Values
page-break-before	auto, always, avoid, left, right
page-break-after	auto, always, avoid, left, right
page-break-inside	auto, avoid
orphans	number (lines)
widows	number (lines)

7.2.3 Page Sizes

```
/* Common page sizes */
@page { size: letter; }      /* 8.5" x 11" */
@page { size: A4; }          /* 210mm x 297mm */
@page { size: legal; }       /* 8.5" x 14" */
@page { size: A3; }          /* 297mm x 420mm */
@page { size: tabloid; }     /* 11" x 17" */

/* With orientation */
@page { size: letter landscape; }
@page { size: A4 portrait; }

/* Custom size */
@page { size: 100mm 150mm; }
```

Thank you for reading!

For more information, visit [MDN Web Docs](#) and [web.dev](#)